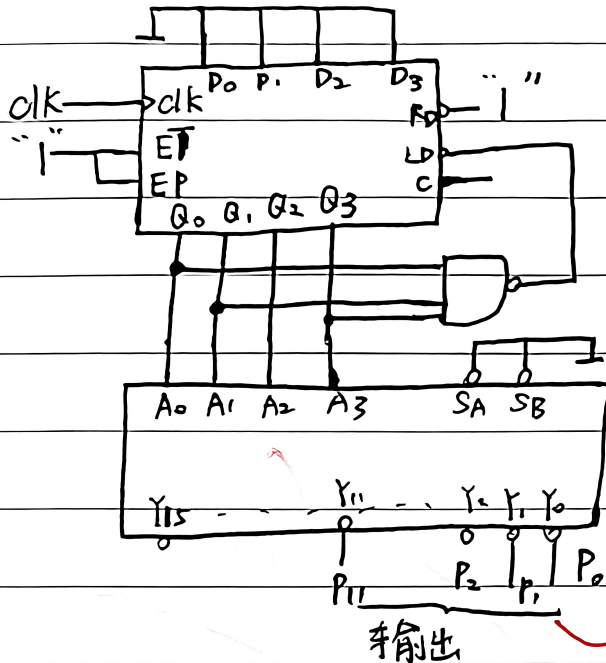
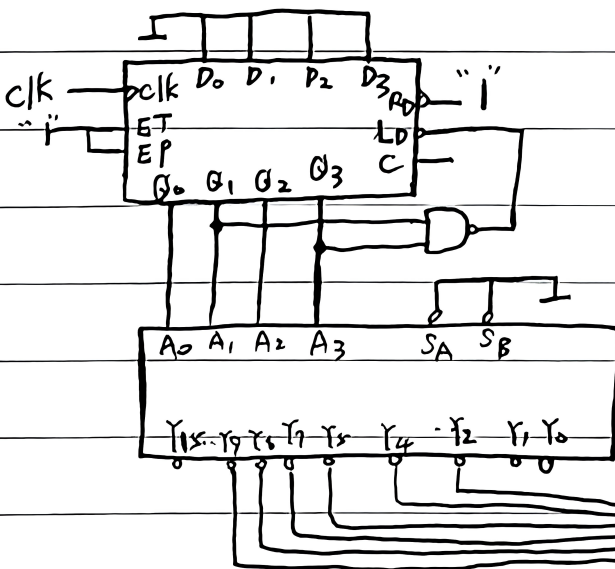


作业8

6.28 只须将计数器改为十二进制, 再接入译码器即可



6.29 共+个状态, 将LS161改为+进制, 再接入译码器译出Y信号



$$Y = Y_2 + Y_4 + Y_5 + Y_9 + Y_7 + Y_8$$

$$= \overline{Y_2} \overline{Y_4} \overline{Y_5} \overline{Y_7} \overline{Y_8} \overline{Y_9}$$

6.34

module motor

input M,

input clk,

output reg [2:0] y

);

parameter S1=3'b001, S2=3'b010, S3=3'b011, S4=3'b100,
S5=3'b101, S6=3'b110;

reg [2:0] state, next_state;

always @(posedge clk) begin

state <= next_state;

end

always @(*) begin

y = state;

case (state)

S1: next_state = M ? S5 : S3;

S2: next_state = M ? S3 : S6;

S3: next_state = M ? S1 : S2;

S4: next_state = M ? S6 : S5;

S5: next_state = M ? S4 : S1;

S6: next_state = M ? S2 : S4;

default: next_state = S1;

endcase

endmodule